

Namrata Yadav

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Education

MS in Computer Engineering, San Diego State University, CA (GPA: 3.4/4.0)

Aug 2024 - Dec 2026

B.Tech. in Electrical & Electronics Engineering, CVR College of Engineering, India (GPA: 3.5/4.0)

Aug 2019 - May 2023

Technical Skills

Analog & Mixed-Signal Circuit Design: Op-amps (two-stage Miller-compensated, 5T OTA), bandgap reference, LDO power management, CTAT/PTAT sensor interface; current-mirror biasing, gm/ID device sizing, compensation and stability, gain/bandwidth/power optimization

Simulation, Characterization & Verification: DC, AC, and transient analysis; loop stability and phase margin, PSRR, noise analysis, PVT corners, Monte Carlo mismatch, reliability aging (MOSRA/URI), temperature characterization (VT, gm, ID extraction); DRC, LVS, ERC, antenna checks, parasitic extraction (PEX), pre-/post-layout correlation, EM/IR-drop

Full-Custom Layout: CMOS/FinFET analog and digital layout, common-centroid and interdigitated placement, device matching, dummy devices, diffusion sharing, Euler-path ordering, guard rings, deep N-well isolation, latch-up prevention, parasitic-aware routing

Digital Design & Physical Implementation: RTL-to-routed-layout, logic synthesis, floorplanning, power planning, placement, CTS, routing, static timing analysis (STA), SDC, setup/hold, WNS/TNS, timing closure, PPA optimization

EDA Tools & Programming: Cadence Virtuoso, Spectre (SPICE), ADE Explorer/Assembler/Maestro, Layout XL; Synopsys Design Compiler, IC Compiler II, PrimeTime, PrimeWave, Custom Compiler, IC Validator, StarRC; Python, TCL, Verilog, MATLAB, UNIX/Linux; SAED 28/32 nm, GPD45

Experience

Research Assistant – Neuromorphic Hardware Co-Design | SDSU

Nov 2025 - Present

- Designed a transistor-level CMOS leaky integrate-and-fire neuron (0.35 μm , 3.2 V) with weighted current-mirror synaptic inputs, subthreshold leak path, inverter-based comparator, and reset feedback; sized devices and mirror ratios ($1\times/5\times/8\times$) against firing threshold, rate, power, and area for a near-linear 8.75 Hz/nA F-I response and 4.86 ms membrane time constant.
- Realized the membrane capacitor as a MOSCAP array (42 $\mu\text{m}/800\text{ nm}$, $m=10$, 52 pF) and built parasitic-aware layout preserving mirror matching and minimizing coupling on the charge-sensitive membrane node; ported measured device parameters into a Python/SpikingJelly LIF model, quantifying the cost of silicon non-idealities on MNIST (94.5% circuit-calibrated vs. 97.4% ideal).

Graduate Teaching Associate | SDSU

Aug 2025 - Present

- Lead circuits, electromagnetism, and optics laboratories for ~ 50 students per semester: instrumentation, measurement, and circuit debugging.

IC Layout Design Intern, Takshila Institute of VLSI Technologies | India

Aug 2023 - Apr 2024

- Delivered full-custom CMOS and FinFET layouts – combinational logic cells and a five-stage ring oscillator – under strict geometry constraints, applying floorplanning, device matching, diffusion sharing, and fin-quantized sizing; closed DRC and LVS on every deliverable through iterative debug of spacing, enclosure, connectivity, device-geometry, and mismatch violations.

Projects

Analog & Mixed-Signal Full-Custom Layout

Cadence Virtuoso, GPD45

- Implemented full-custom layouts for 10+ analog and mixed-signal blocks — two-stage op-amp, bandgap reference (5×5 common-centroid BJT array), LDO, level shifter, Schmitt trigger, PLL — using common-centroid and interdigitated placement, device symmetry, dummy devices, guard rings, and deep N-well isolation to hold matching and block substrate noise.
- Designed a 1.8 V-to-1.2 V LDO (5T OTA, thick-oxide PMOS pass device, enable-controlled bias) reaching 1.192 V output and 149.8 mV dropout, routing the 96 μm multi-finger pass device with wide-metal, multi-via power rails for IR-drop and electromigration headroom; verified lifetime drift through Spectre MOSRA/URI aging analysis.
- Delivered DRC/LVS/ERC-clean hierarchical layouts and closed metal-density and fill requirements; ran PEX and pre-/post-layout comparison, resolving connectivity, matching, routing, and enclosure violations.

Digital CMOS Standard-Cell Library & Hierarchical Custom Layout

Synopsys Custom Compiler, SAED 28/32 nm

- Authored an 11-track CMOS standard-cell library (INV, BUF, NAND2, NOR2, AND2, OR2, XOR2, XNOR2), applying Euler-path ordering to maximize diffusion sharing and minimize cell width, with fixed poly pitch, uniform cell height, aligned VDD/VSS rails, and grid-aligned M1 pin access for P&R; built muxes, flip-flops, counters, and 8:1/1:8 SerDes from it.
- Built a 6T SRAM bitcell and array with shared supply rails, word-line distribution, differential bit-lines, and latch-up-immune well taps; cleared DRC/LVS/ERC and antenna checks in IC Validator, extracted with StarRC, and verified timing in PrimeWave.

Physical Synthesis & Timing Closure of a 32-bit Pipelined ALU

Synopsys SAED 28 nm

- Drove a 32-bit pipelined ALU from RTL to routed layout in Design Compiler and IC Compiler II — synthesis, floorplanning and utilization, power-ring and strap planning, placement, CTS, and detail routing.
- Closed post-route timing by authoring SDC constraints and running PrimeTime STA on critical paths, driving WNS/TNS to closure and clearing clock-skew, setup/hold, transition, capacitance, and fanout violations through cell sizing, buffer insertion, and placement/routing optimization; automated timing, area, power, and violation reporting in TCL/Python.

CMOS Temperature Sensor – Device Characterization & CTAT Design

Cadence Virtuoso, GPD45

- Developed a CTAT/PTAT CMOS temperature-sensing architecture with a Miller-compensated two-stage op-amp, optimizing sizing, bias current, and compensation for sensitivity, gain, stability, and power.
- Characterized -25°C to 125°C in Spectre/Maestro, extracting V_T , g_m , and I_D trends and selecting a $2\mu\text{A}$ CTAT bias point holding monotonic operation across the full 150°C range at $-0.201\text{ mV}/^{\circ}\text{C}$, with post-layout PVT, Monte Carlo mismatch, and aging analysis.